MIL-R-46452A (MU)

31 MARCH 1965

SUPERSEDING
MIL-R-46452 (Ord)
30 DECEMBER 1960

MILITARY SPECIFICATION

RELAY, M6

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions, and special requirements not covered by the drawings, for the loading, assembling and packing for one type of relay designated as Relay, M6.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548347 — Box, Set-up, Packing, Ammunition, Outer for Relays or Primers, Assembly, Details, Packing and Marking.

7548348 — Box, Packing, Ammunition for Relays, M1, M2 or M6, Assembly, Details, Packing and Marking.

7548389 — Box, Set-up, Packing, Ammunition, Inner

FSC 1375

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for Relays or Primers, Assembly, Details, Packing and Marking.

8844539 — Relay, M6, Loading Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8844539 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Relay. The relay shall comply with all requirements specified on Drawing (dwg.) 8844539, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. The moisture content of the lead azide used in loading the relay shall not exceed 0.50 percent when tested as specified in 4.3.1.

3.4 Workmanship. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specification referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be sub-

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stantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of relays produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Sublots shall be submitted in accordance with Standard MIL–STD–105 or MIL–STD–1235, as applicable. Each lot shall contain—

(a) Parts bearing the same lot interfix number from one manufacturer.

(b) Lead azide from not more than one lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except the inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL or 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Relay assembly (see dwg. 8844539).*

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	**None defined.**		
Major:	**AQL 0.65 percent**		
101.	**Outside diameter, maximum**	**Gage**	**01001**
102.	**Crimp improper**	**Gage**	**01002**
103.	**Cup damaged**	**Visual**	**01003**
104.	**Charge missing**	**Visual**	**01004**
Minor:	**AQL 0.65 percent**		
201.	**Evidence of poor workmanship (see 3.4)**	**Visual**	**01005**

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4.2.2.2 *Box, Set-up, Inner (see dwg. 7548389).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Contents loose	Manual	02001
Minor:	AQL 1.00 percent		
201.	Label missing	Visual	02002
202.	Marking incorrect, incomplete, or illegible	Visual	02003

4.2.2.3 *Box, set-up, packing, ammunition, outer (see dwg. 7548348).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined:		
Major:	AQL 0.65 percent		
101.	Contents loose	Manual	03001
102.	Sealing compound incomplete or inadequate	Visual	03002
103.	Waterproof seal broken	Visual	03003
Minor:	AQL 1.00 percent		
201.	Label missing	Visual	03004
202.	Marking incorrect, incomplete or illegible	Visual	03005

4.2.2.4 *Box, packing (see dwg. 7548348).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Strapping missing, broken or loose	Visual-Manual	04001
102.	Box damaged to the extent that contents are exposed or liable to become exposed	Visual	04002
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	04003
202.	Strapping improperly assembled	Visual	04004
203.	Marking incorrect, incomplete or illegible	Visual	04005

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4.2.3 *Testing.*

4.2.3.1 *Moisture content of lead azide (see 3.3) Major defect — Code No. 05001.* The moisture content of each lot shall be determined at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.1. If the moisture content is in excess of the percentage specified, and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.1. If loading has begun, action on the unloaded lead azide shall be as specified above and all relays loaded with the questionable lead azide shall be rejected.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Number IEL–8844539 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of moisture content of lead azide.* A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrofluoroethylene 2½ inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to a moisture dish and the contents and the dish shall be transferred to an oven and dried at 95° Centigrade (°C) ± 2.°C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrofluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to the percentage of moisture content in the lead azide to determine compliance with 3.3 as follows:

$$\text{Percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the sample before drying

B = the weight of the sample after drying

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level C.* The relay shall be packaged and marked in accordance with dwgs. 7548347 and 7548389.

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5.2 Packing.

5.2.1 *Level C.* The packaged relays shall be packed and marked in accordance with dwg. 7548348.

5.3 Data cards. Data card information shall be prepared for each lot as specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The 5-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and nonmandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, de-

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scriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within the month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

Custodian:

Army—MU

Preparing activity:

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